

1.6.1 Data Visualization

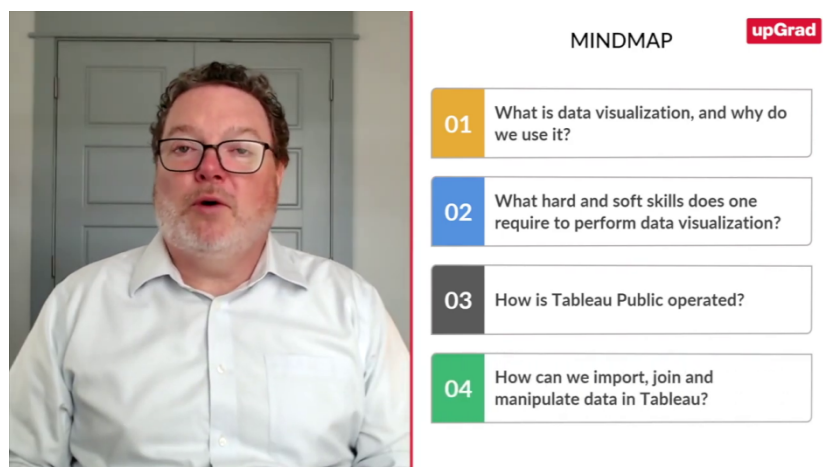


Sean Oslin

Hello, I'm Sean Oslin, I'm a data scientist in the United States and I will be teaching the introductory course in Tableau. I'm a self-taught in Tableau. I spent about three years sort of digging into it and learning how to do it through school projects, online projects and on my own, and I hope to convey to you what I have learned, both in theory and in practical applications and in terms of working with Tableau.

A little bit more about me. I have degrees from the University of Michigan and the University of Washington. I spent about 25 years working in health administration and transitioned over to data science about four years ago.

Currently I work with my company Booz Allen Hamilton as a contractor with the Centres for Disease Control and Prevention in the United States on their Covid-19 response. I would encourage you to follow Booz Allen Hamilton on LinkedIn because we are always hiring data scientists.



MINDMAP

- 01 What is data visualization, and why do we use it?
- 02 What hard and soft skills does one require to perform data visualization?
- 03 How is Tableau Public operated?
- 04 How can we import, join and manipulate data in Tableau?

The module mind map will include: first we'll discuss data visualization and why we do it.

Second, we will talk about the hard and soft skills you need to make good data visualizations. Third, we'll talk about the basics of operating the Tableau Public website. Next, we'll talk about how to import, join and manipulate data in Tableau. After that, we'll work on creating common graphs in Tableau.



WHAT IS DATA
VISUALIZATION?

upGrad

Sean Oslin

So, just what is data visualization?



WHAT IS DATA
VISUALIZATION?

upGrad

- 01 It is an interdisciplinary field
- 02 It is a way to represent data graphically
- 03 It makes big, complex data understandable and actionable
- 04 It involves storytelling using data

Well first, it's an interdisciplinary field and by that I mean there are you know people from working in health care, people working in the sciences, people working in public policy who all are creating visualizations and they're also helping push the field forward by creating innovative new designs and learning new techniques.

So there's really not a field in the world nowadays that doesn't use visualization in some way. It just communicates better than just simply talking to people or presenting them with numbers. Data visualization also is, by its very name presents data in a graphic format. Now that doesn't mean everything has to be a graph.

You can also use tables too. A matter of fact, there's often I have found in my experience that employers want to have both the graph that shows things as well as the data table, so they can look at the data as well.

Another key point about data visualization is that it's taking complex data and trying to create it, recreate it in a format to adjust to transform it in such a way that we can actually generate information from that and from the

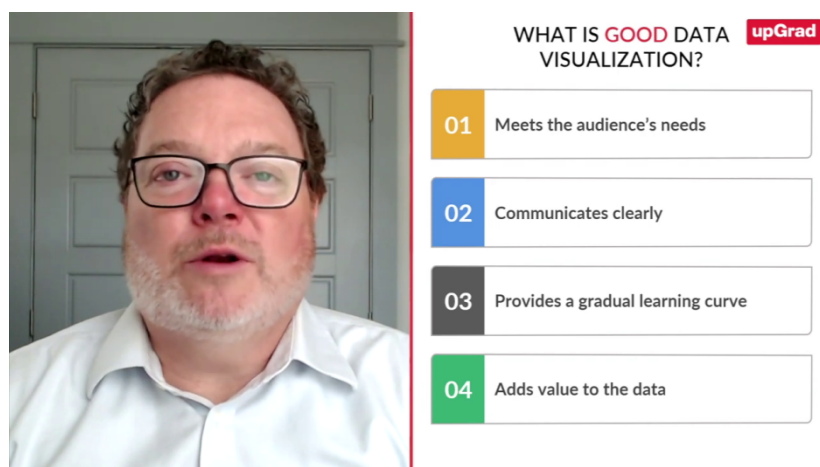
information we can create actionable tasks that we can do that will either move our business forward or address a problem or something else.

And last but not least, data visualization is about storytelling. We have all this data up here and then we need to put it together with other data and then we need to say this is what it's telling us. This is what we want to communicate. This is what the actions we need to take.

So it's not enough to just create pictures with the data we need to actually make it actionable. We need to be able to say, we've come from here, we're going here and that's really, you know the meat of data visualization is about. What is the story, we're trying to tell, what are the actions we are trying to take?



So what is good data visualization?



There's a lot of bad data visualization out there and I'll actually show you some examples of that. But you know the probably the key-key point of data visualization, good data visualization is it meets the needs of your audience and I'm going to say this over and over as we go through the course.

It's like you really need to understand who your audience is and what they're looking for. So you really should ask for feedback during the development process. The last thing you want to do is could spend months working on the data visualization and then you get and roll it out and people say that's not what we're looking for.

The best thing to do is to talk to them, talk to you the people in charge as much as you can, if you can get access to them. As you go and so you're actually meeting their needs. Another thing is it's not about you, it's about your audience. So don't be too clever. Don't go off and create some nifty note graph that looks really cool, but is not that easy to read.

Just make sure that your audience really can understand it quickly because they're not spending as much time with the data as you are. They need to really understand what it is that you're trying to communicate to them. Again it's about storytelling.

Good data visualization will also communicate with a minimum amount of complexity. Now you're already taking all this raw data. You know massive amounts of raw data and you need to condense it down, but you need to make sure that you're condensing it down far enough that people can then make some understanding of it. You can't make it too small either, you can't aggregate the data until it says nothing.

So it's kind of a balance and if you're not you if you don't understand your subject, you're not a subject matter expert, you're going to need to work with people who do understand the data as you go. In my experience, I have worked on several projects where, honestly I did not understand what it was that I was working on.

I was not a subject matter expert in the area, and I had to continually ask for help from people saying is this meeting your needs? Are you getting what you want? I don't understand what this means is this right? Do I need to join this variable with this variable? It's like, so it was like all right.

Is this too much detail or is it not enough detail and you know you just have to work with people, because they often don't even know what it is that they want. So you have to kind of lead them through the process. Also, keep in mind that, if you're presenting a graphic at a meeting that your audience is going to need to learn how to read it, there's going to be a learning curve involved for them.

So again, you need to make it as simple as possible for them to comprehend the data rapidly and if you need to have more complex backup material or things where you're drilled down into the data, you can have that as supplemental material. But again you need to think of it, if it's being presented at a meeting for the first time.

Think of it like as a summary. Another thing that good data visualization does is it helps you explain concepts that are too complicated to show in words. Maybe you can actually demonstrate them to a graph and the audience will comprehend it as many times as quickly as they would, if you were to just speak to them or to put it down on the printed page as text.

Finally, good data visualization is ethical. You are telling the truth, and it's very-very important that that you keep it in mind when you're building a data visualisation, it's like, is this an actual representation of the truth? Am I skewing the data to make it look like, it's something that it's not.

It's like and honestly, that's very easy to do and it does happen and I'll give you some examples of this. But you need to think about your personal integrity when you're building something to make sure that what your work, what you're showing is sort of a representation of the truth.



So we've talked about good data visualization. Let's talk about some bad data visualization.

BAD DATA VISUALIZATION: EXAMPLE



This graph is what I would call generously to be generous about it, misleading, if not outright lying. In particular, what you should look at is the y-axis and look at how the intervals change but between them you go from 1 to 30 and then things start jumping, but to different extents like up to 50, I believe.

But the lines are all staying equal distance, the horizontal lines. So I'm not quite certain what they're trying to get across here. But you know this is just wrong and I really just want to let you know that you should be real critical consumers of data visualization and, as you get better at it, you'll just start seeing everybody's mistakes.

BAD DATA VISUALIZATION: EXAMPLE

upGrad



Here is another example of a misleading graph, the one to start out with the one on the left. If you look at it, the scale doesn't start at zero. It starts at 30. So what that does is it completely exaggerates the difference between the two. When you look at the corrected graph on the right, you can see that the differences are not so extreme.

And this comes back to the fact that people will they look at the visualization aspect of it more than they look at the numbers. So it looks in the one chart on the left. It looks like the differences are extreme, but once you start, you know once you correct it and you know people won't even notice, they won't even look at the scale, but once you correct it, you know you suddenly get a better sense of what the truth is.

Just keep in mind that your audience is not necessarily data literate. They don't necessarily know how to read charts and graphs. So it's your duty to make sure that they are able to do it without being misled.



upGrad

How are data visualizations used?



There are four main ways. First, descriptive analytics. This tells us what happened. We can look at our past financial year to see: oh well, we made a profit or we made a loss. So it's just very-very simple. The most common way to use data visualization. Second, is diagnostic analytics.

This helps us explain why something happened. If we missed our target for something, we can go back and look it to see what fed into that? What was the sort of the key elements that caused something to happen? So we can go in and just find out what the issues are. Third is predictive analytics.

So what we want to do now is look into the future. We'll say: okay, we're we did well this year. This is what we expect to happen with the stock market. This is what we expect to happen with adding new customers, expanding our base of operations and this is where we think we're going to be 6 months, 12 months, 24 months into the future.

Well, of course, the further into the future you predict the more problems you have with your prediction, the more noise ends up in there and the less precise your prediction is. But at least you're trying to see where you're going to go or even do some sort of analysis where we could look under these conditions, this is where we think we'll be, but if this happens then this is where we think we'll be.

Finally, we'll talk about prescriptive analytics. This is sort of a you say you are running a machine and you have certain thresholds for the operation, which was which defines what is normal. But when the machine suddenly goes outside the you know the normal range say: you're getting more chaff from something that you're milling.

Maybe it's time for preventive maintenance. So prescriptive analytics was, could really come in and say: hey look, things are not normal anymore. We need to do something about it, like get our machine serviced.



So what is Tableau?



WHAT IS TABLEAU?

- 01 It is a database
- 02 It is a business intelligence (BI) tool
- 03 It is a graphical presentation software
- 04 It can be used to monitor existing operations

First and foremost Tableau is a database. It operates under its own proprietary version of SQL. So just keep that in mind that is a relational database. It works with columnar data, just like any other database that you might work with. Second, Tableau is a business intelligence tool.

So Tableau is also graphical presentation software, so its sole purpose is really to take data, raw data, aggregate it and then present it in a way that is understandable to your audience, a way that you can tell stories to your audience. So Tableau is also a way to monitor existing operations.

So dashboards become very popular nowadays to be able to manage your inventory or monitor your inventory, monitor your financials, your HR, your hiring, your payroll, all those things. So it's are tied in with business intelligence tools like Tableau.



WHAT IS TABLEAU?

upGrad

- 03 It is a graphical presentation software
- 04 It can be used to monitor existing operations
- 05 It can be used to discover new business opportunities
- 06 It is relatively non-technical

Tableau can also help you discover new business opportunities through going through and analysing the data. You're often able to say: hey, here is a market niche we have not looked at. We think there may be some real growth opportunity here and maybe we should go in and try to exploit that, and you know Tableau is a relatively non-technical way of doing business intelligence.

You don't need to learn Python to use Tableau, but honestly, it really helps to have some SQL skill and to have some other sort of basic-basic computer programming skills to do stuff on the back end. The more you know about programming, the cooler you can make things in Tableau or the more complex the business questions you can answer in Tableau.



upGrad

So there are many different versions of Tableau and the one we're going to work with is Tableau Public.



WHAT IS TABLEAU PUBLIC?

- 01 It is a free software similar to Tableau Corporate
- 02 It is a website and desktop app
- 03 It is semi-private
- 04 It offers storage capabilities (for storing one's public portfolio)

So what is Tableau Public? First of all and most importantly, it's free, which is very handy because the Tableau Corporate is a very expensive proposition and you often have to convince your employers that you need it in order to buy it. Tableau Public contains most of the capabilities that the Tableau Corporate edition contains.

There are some limitations, but for particularly for people who are new to Tableau, it has everything you need to go in and learn how to make things, to make visualisations. Sometimes we hear people talk about Tableau Public, we're trying to figure out: is it software or is it a website? And the answer is it's both.

Tableau Public is the free version of Tableau that you use when you're a student and Tableau Public is actually where you store all of your visualizations. Tableau Public is as we said is a place you store your visualizations. Tableau Public is only semi-private, you can go in and most the default when you store things at Tableau Public is it's going to have it's going to be public, literally public.

That means anybody can go in and see what you're working on. You actually have to go in and change the settings to either be have your default, that is not going to be public or you can manually hide visualizations that you're working on. I do that frequently when I'm working on drafts of things.

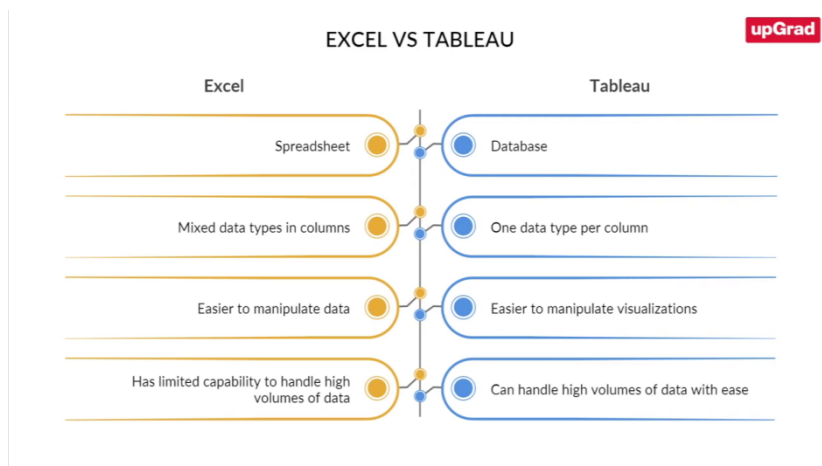
So if since it's public or semi-public or only semi-private, you might think wow is this a good place if I'm working on a budget for you know for my company? Is this a good place to store the budget? And you're like the answer is absolutely not. Do not put any proprietary material, anything that's private.

So no financial, no HR, no business intelligence information at Tableau Public, because people can go in and find it. They can google you and they can google your visualizations and they can go in and see it, even if you are going in and hiding your visualization, making it non-public, I still would not recommend you ever store anything related to your business at Tableau Public.

You need to get the Tableau Corporate edition and be able to store it either on your local computer or on a server. Finally and this is probably the best, one of the best ways to understand Tableau Public. Tableau Public is sort of like GitHub for your code. You can't store visualizations at GitHub, at least, if you're, using Tableau Public, but you can store them at your Tableau Public account.



So, if your boss comes in and says why do we need Tableau? He goes, isn't it just like Excel and your answer is no, it is not just like Excel.



The slide will actually show some of the big differences between it. But let me talk about a few of those. First of all, Excel is a spreadsheet. The focus is on the cell level. The data can readily be changed and a lot of companies consider that to be a security issue, I mean, if you can go in and monkey with the budget, then that's a very serious issue.

Tableau, on the other hand is a database and you know with databases, there are specific ways you have to go in and change things. And with as with Tableau, usually the data has to be changed on the feed coming into Tableau. So there's a lot more control over it.

There's a lot more data integrity with Tableau. Also with Excel, you can have multiple data types in the same column. You can also have, so you can have like texts and you can have numbers, all those things in the same column, you can have formatting, you can have all sorts of things to make things look pretty for your end user. Well with Tableau, it's the other way around.

The data has to come in raw and then you can go in and create all sorts of amazing sort of graphics based on the data, far more than you can do with Excel. So Excel is really an exceptional way to take the raw data and readily make it into a limited number of different tables or graphs within, while manipulating the raw data.

In Tableau, it's the other way around, the raw data stays put, you don't touch the raw data, but when the data is brought in, then you can manipulate it. So the data is preserved and then you can create do all sorts of manipulations, pivoting the data, creating new variables, all sorts of things you can do, which you can do to a certain extent in Excel, but you've just much greater capability in Tableau.

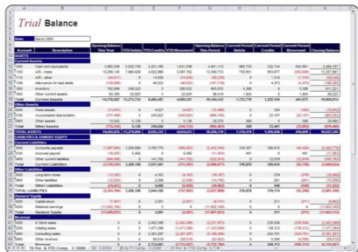
So there's also a big issue with size. Tableau has basically unlimited capabilities, since it works with tables you can have, you can have 50 different tables that are all linked together and feeding into your into your visualization. With Excel, you're really kind of more or less limited to one spreadsheet.

You can link to others as well, but you know it's not as easy to do and you have an absolute row limit of a little over a million rows right now. There may be more in the future, but I'll say when you get that much data in Excel, the performance plummets.

You really cannot, you cannot operate Excel with that much data, especially if you start putting pivot tables in there or you start doing other sorts of really complex calculations. What you know would take seconds to do in Tableau, can take minutes to do in Excel.

HUMAN-READABLE VS MACHINE-READABLE DATA

Human-Readable Data



Machine-Readable Data

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	
1	CA-2015-052056	08-12-2015	12-12-2015	Second Class	CG-12520
2	CA-2015-052056	08-12-2015	12-12-2015	Second Class	CG-12520
3	CA-2015-128688	12-06-2015	16-06-2015	Second Class	DV-12045
4	US-2015-108969	11-01-2015	18-01-2015	Standard Class	SO-20335
5	US-2015-108969	11-01-2015	18-01-2015	Standard Class	SO-20335
6	CA-2014-125822	09-09-2014	14-09-2014	Standard Class	BH-12170
7	CA-2014-125822	09-09-2014	14-09-2014	Standard Class	BH-12170
8	CA-2014-125822	09-09-2014	14-09-2014	Standard Class	BH-12170
9	CA-2014-175017	09-09-2014	14-09-2014	Standard Class	BH-12170



Now, let's talk a little bit about machine readable and human readable data. Now the example on the left is really of a budget and this is how you would expect to read a budget and it's very, it's got formatting in it. It's got totals, even subtotals, it has maybe some calculated numbers and it's laid out where it's one month after another going to the right.

Now that is that's perfectly great. That's a perfect thing for Excel to do, but you can't take the data from in this format and turn it into and upload it into Tableau. This is, as I said, human readable data. What you need is machine readable data, which is the same sort of format you would use for SQL or for any other database. So sometimes it's called human readable is called wide data and machine readable is considered long data.

Now I think really the big difference is that, with machine readable data is you only have one data type per column. You only have one. The column has a name, and everything in that column is related to the you know whatever that column name is. So if it's product name, everything in that column is gonna be a product name. If it's give you sales date, every date is gonna have its own line.

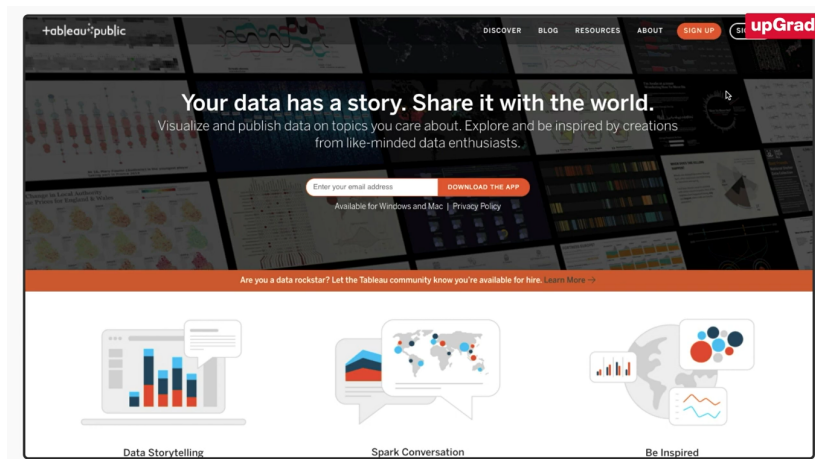
Every and every product purchase on those dates is going to have its own line. So, instead of having data that's you know could be like really-really wide showing all of these things that are going on. Again, it's going to be generally the data will be narrower when much longer to account for the way that you need to upload the data into Tableau to make it readable.



Now, I've often been asked to replicate data like I want you to create the spreadsheet, but I want you to do it in Tableau. I said okay and they were like well, I said: well, what's the data? They said well just take the budget and do it that way, and I was like great.

So then you have to go through and reformat all of the data to get it into a long format. So, I always try to like say: can I work with the raw data? Because any sort of data coming out of any sort of financial database is going to be in a long format.

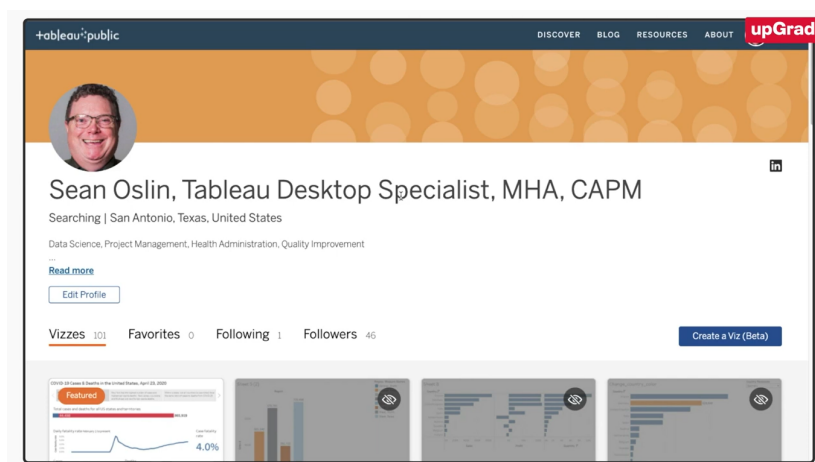
It's going to be machine readable and from that you can, I can upload into Tableau and I can create visualizations. But to take you know, to take somebody's Excel spreadsheet and then create something just like it from that data in Tableau is going to require a tremendous amount of data reformatting in Excel before you can bring it into Tableau.



Hello, let's get started with tableau. First thing we're going to do is download the app, and we're going to do that from the tableau public website. Now, one of the first things people have asked me about tableau public, is it software or is it a website for storing your visualisations? And the answer is yes or really both. You download the program from here the public version of tableau, and at the same time, you will be storing all of your visualisations here at this website. So, to begin, I want you to put your email address into the white box and then download tableau public.

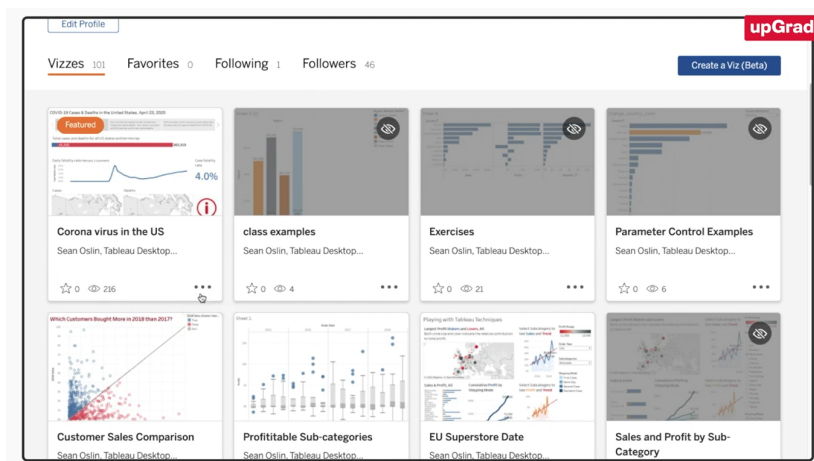
Now, there are quite a few steps you're going to need to follow to get the app down like you would any other app, so just go ahead and do that, pause the video until you're done and then start the video again once you're ready. At this point, I assume that you have all been able to download and open tableau. So now, we're going to sign you up for your tableau public account. Going back to the tableau webpage, go and click in sign up and go ahead and fill out all the different questions here that you need to create an account.

You do need to use your real name. Remember, you want people to find you. This is like GitHub. This is your public portfolio. This is you want people to be able to go in and Google, you will find you. Use an email address that nobody will see it, but you do need to have an email address and where you can link everything. Put in your password, mark that you've read the terms of service and click confirm that you're not a robot. Pause the video. Once you're done with this, you'll come back, and we'll go into tableau.



I assume at this point that you all have created a tableau public account. So now, we're going to go sign in. So, I'll sign into my account, and this is what my account looks like from the inside. This is my view of it. This is not be what you would see if you were to Google me, which is exactly what I want you to do right now. Go to a browser and search for my first and last name, Sean Oslin and then write tableau public after that and what should come up should be either this main page or one of my visualisations that I have created. So, pause the video and when you're done with that, come on back.

At this point, I'm assuming that you're all looking at one of my visualisations or aap my main page. Try to get to the main page, and then I'm going to walk you through some of the things that you can find in tableau public. The first thing you need to look at is all the information that's stored here. My name, has my certification, says where I am, says what my interests are. You can write more to your profile if you'd like. This is very important. This is how people can get to know a little bit about you, kind of like on LinkedIn, but not that many, not as nearly as many people will look at this, but it is kind of important to kind to fill this out.



Then, the next thing you should look at is the number of visits. Now, when I'm inside my account, it says I have 101 visits, and that is not what you're going to be seeing, your number will be probably about half that, because I have a lot of hidden visits in here, things that are grade out you can see, and they have the eye with a line through it. These things are hidden on your profile, nobody, but you can see them, and I frequently do this with drafts of things. I don't want things to become public until I've fully developed it. So, I just hide it. You can also change the default, so everything that you create is hidden and if things only become public, when you specify that, and I think that's a pretty good idea, I would recommend doing that.

One of the things that people ask as well is like say I'm waiting for my company to get me the tableau corporate division to get the tableau corporate application. Can I just go ahead and use tableau public, and you know, work on my work projects and store everything at tableau public, and I would just say do not do that. While you may be able to hide things, things are only sort of semi-private. If you were to go and put the link, this actual link to one of the one of the visualisations, if you were to email that to somebody, it doesn't matter whether it's hidden or not, they will be able to access it.

So, I would just be very careful about putting any sort of confidential business information or personal information into a visualisation, because accidents do happen and the last thing you want to do is get fired because you release confidential information. So really, to sum up, your tableau public account, since it is your public portfolio, it's what

faces the public should only include things like your school projects, things that tutorials you followed from online courses or projects that you've done like I did this coronavirus visualisation last year and while I was job hunting.

So, I was developing my skills and I was building up my portfolio. But you can see that I have, you know, a lot here and people go in and look at this. They're really curious about what I have in my portfolio, and this gets you jobs. So, at this point, we'll stop here, and we'll move on to opening up and connecting to data in tableau.



So some of you may decide that you want to specialise in data visualization as part of your data science career. It can be a lot of fun to do that and there's certainly plenty of jobs out there for people who are interested in this level of specialisation. So the question is like: how do you actually get yourself hired?

Well, you know like with any like with artists or with other sort of professions, you have to have a portfolio to work with and something that you can show potential employers, how you've done your work? What you've learned, what you're capable of?



BUILDING TABLEAU PORTFOLIO

- 01** YouTube videos, class material, and other sources
- 02** Personal projects (e.g., Sean's COVID-19 dashboard)
- 03** Fun competitions such as <https://www.makeovermonday.co.uk/>

So the best way to do that is to actually go in and then just do all sorts of visualizations. So you can go and look at YouTube videos. There's all kinds of how-to videos. How do you build a Sankey chart and you can go in and you can just find it like wow that looks really cool. Usually they have the data there.

You could download it and you can create visualizations, learn how to create new visualizations from you know a 10-15 minute video on YouTube, and then you save it to Tableau Public and when you've cleaned it all up, make it look professional. It's like wow. Okay, then you got something you can add to your portfolio.

That is your portfolio. Tableau Public is your portfolio. You can also do the same thing with your class projects here, they will all be part of your portfolio and you could take classes from Udemy or other online schools. I have you know tremendous number of visualizations which I've done in class, in online classes and I've taken a lot of them.

So I would you know just make sure, though, when you know you may just do it very rapidly in one of these classes, or one of these videos, clean it up, make it look professional, do all the sort of best practice things that we'll talk about to these things and you put it up there and that is your representation of your portfolio for Tableau.

So another way you can build your portfolio is by doing personal projects. The one listed here is a COVID-19 project that I was working on my own, while I was job hunting a while back at the beginning of the epidemic. So it was, it you know was I was getting the data daily and then I was updating it every day with the new data, I would just go in and connect to the new data set and it would update it daily.

And this was in my Tableau Public portfolio. It's something that employers could see and you know just it helped me you know when I was looking for a job. Probably because I had so many of them. I had you know like 50 visualizations in there that I had built up in my portfolio and you know that was just a big selling point when I was talking to employers.

Finally, in terms of doing your building your portfolio, you could do some of these really fun things like Makeover Monday and which is a, it's a website and it's a book and it's a I don't know if you really call it competition, maybe it is but they take a data set every week and then they go and redo it to make it you know, take somebody's visualization, then redo somebody's visualization or they just work with raw data and then they just create new visualizations based on it and people can post what they've done and people can vote on them or comment on them and it's a great way to learn, it's a great way to improve your skills.

One company I worked for, they had, they did a Makeover Monday once a month and so everyone got the data and then they had a week where they could work with the data and then they came back and everyone presented what they had done in a big meeting.

People wanted to show up there, so they could so they could say: Oh, this is how I did you know this worked with this data set and what I came up with. So it's a great way to just to learn new skills and to actually make some connections within the Tableau community.



If you're really interested in data visualization as a career, I really would encourage you to look at Tableau's certifications. And these are really helpful in terms of showing to employers that you have the job skills to meet either sort of beginning to intermediate to all the way up through advanced or very advanced Tableau skills that you can that they can use in in their workplace.



So I have the Specialist certification, which is the first one on the list and you know that's really helpful. I have that you know on my resume. I have it in my signature line of my email, along with the link to my Tableau Public account and also my GitHub account.

So I you know, I really would recommend that you look at it. They often will have a sale on it because it still costs like you know, minimum of a hundred dollars. But it is really a great thing to show that you've actually had some of the skills involved and they have there's quite a few trainings involved with it or you can do through Tableau. You might be able to find them less expensive through Udemy or something like that, but you can take a look.

But there are actually specific courses out there designed to help you pass the certification. I would start out with the specialist and then work up from there. I don't know much about the data analyst one, that's brand new and I haven't had a chance to explore that one yet.

Disclaimer: All content and material on the upGrad website is copyrighted material, belonging to either upGrad or its bona fide contributors, and is purely for the dissemination of education. You are permitted to access, print, and download extracts from this site purely for your own education only and on the following basis:

- You can download this document from the website for self-use only.
- Any copies of this document, in part or full, saved to disc or to any other storage medium, may be used for subsequent, self-viewing purposes or to print an individual extract or copy for non-commercial personal use only.
- Any further dissemination, distribution, reproduction, and copying of the content of the document herein, or the uploading thereof on other websites, or use of the content for any other commercial/unauthorized purposes in any way that could infringe the intellectual property rights of upGrad or its contributors is strictly prohibited.
- No graphics, images, or photographs from any accompanying text in this document will be used separately for unauthorized purposes.
- No material in this document will be modified, adapted, or altered in any way.
- No part of this document or upGrad content may be reproduced or stored on any other website or included in any public or private electronic retrieval system or service without upGrad's prior written permission.
- Any rights not expressly granted in these terms are reserved.